

Demo Recording Script

Zerve Upgrade Prediction Notebook

How to use this PDF

Use this as presenter notes while recording. Each segment tells you which notebook cells to open, which output or figure to point at, what to say, and what conclusion the audience should take away. The flow excludes the Streamlit app and focuses only on the notebook project.

- Suggested length: 6 to 9 minutes.
- Start from the top of t1.ipynb and scroll only to the referenced cells.
- Spend the most time on leakage safety, model results, SHAP, funnel, and insights.
- Do not explain every line of code. Explain the decision and the output.

Demo Flow Overview

Use this as the recording order.

Time	Cells	Segment	Output to show
0:00-0:45	Cells 0-5	Dataset and target	Show shape: 3,509,628 rows x 83 columns.
0:45-1:30	Cells 6-24	EDA and leakage scan	Explain missingness and high-risk event filtering.
1:30-2:30	Cells 26-31	Feature engineering	Show snapshot cutoff and final 16,496 x 44 feature table.
2:30-3:45	Cells 32-36	Upgrade model	Show ROC-AUC, PR-AUC, and feature importance.
3:45-4:45	Cells 37-40	Interpretability and tiers	Show SHAP and risk-tier outputs.
4:45-5:55	Cells 41-46	Funnel	Show deterministic stages, funnel chart, transitions, time-to-stage.
5:55-7:20	Cells 47-54	Behavioral insights	Explain AI power users, deployment intent, early activation.
7:20-8:30	Cells 55-63	Supporting visuals and wrap	Show trend, retention, segmentation, distribution, checklist.

1. Project Setup and Dataset

Notebook cells: 0-5

Output to point at

Cell 4 output: (3509628, 83). Cell 5 output: column list with person_id, timestamp, event, person properties, AI fields, credits, deployment and product activity fields.

What to say

I start by loading the Zerve telemetry dataset. The key point is that this is event-level product data, not one row per user. The target event is subscription_upgraded, so the challenge is to turn this raw event stream into a user-level prediction problem.

What can be inferred

The dataset is large enough to build behavioral features, but the target is rare. That means accuracy is not a useful metric, and the pipeline needs to be careful about leakage.

Transition line

Next I check data quality and identify which event types are safe to use before modeling.

2. EDA and Leakage Risk

Notebook cells: 6-24

Output to point at

Cells 8-11: null coverage summary. Cell 22: top upgrade/commercial event types. Cell 24: leakage-risk categories.

What to say

This section shows that many columns are sparse because they only apply to certain event types. That is expected for telemetry. More importantly, I separate normal behavioral signals from commercial or checkout-adjacent signals that could leak the answer.

What can be inferred

Columns and events close to billing, checkout, subscription, credits, or the upgrade target should not be used naively. The model must predict upgrade intent before the user upgrades, not recognize upgrade behavior after it has happened.

Transition line

With leakage risks identified, I move into the most important part: building the user snapshot and feature table.

3. Leakage-Safe Feature Engineering

Notebook cells: 26-31

Output to point at

Cell 27: total users, upgraders, conversion rate. Cell 28: safe events retained and dropped. Cell 31: final feature table, 16,496 users x 44 columns.

What to say

Here I define a snapshot time for every user. For upgraders, the snapshot is their first subscription_upgraded event. For non-upgraders, it is their last observed event. Every feature is computed only from events before that snapshot.

What can be inferred

This makes the prediction setup realistic. The final table has 43 predictive features plus the label, including activity volume, recent activity, AI usage, credit pressure, deployment, sessions, and persona attributes.

Transition line

Now that the feature table is safe, I can train and evaluate the upgrade prediction model.

4. Upgrade Prediction Model

Notebook cells: 32-36

Output to point at

Cell 33: X shape and positive rate. Cell 34: fold metrics and CV summary. Cell 35: Fig 1 model performance. Cell 36: Fig 2 feature importance.

What to say

The model is XGBoost with stratified 5-fold cross-validation. I use ROC-AUC and PR-AUC because the class is imbalanced. The key result is around 0.78 ROC-AUC, which shows meaningful separation between likely upgraders and non-upgraders.

What can be inferred

The feature importance output shows that upgrade intent is most associated with AI token usage, recent events, credit pressure, total activity, AI generation counts, and product-building behavior.

Transition line

A model score is useful, but for a product team we also need to explain why users score high and how to act on the score.

5. Interpretability and Risk Tiers

Notebook cells: 37-40

Output to point at

Cell 38: Fig 9a SHAP beeswarm and Fig 9b SHAP bar. Cell 40: risk-tier table and Fig 10.

What to say

SHAP explains the model at both the user level and global level. The beeswarm shows whether high or low values push individual predictions up or down. The bar chart shows the strongest overall drivers.

What can be inferred

The risk-tier output turns probabilities into product actions. Very High users have the strongest conversion concentration and deserve high-touch outreach, while Low and Moderate users are better served with scalable nudges.

Transition line

After modeling individual upgrade probability, I switch to the second challenge: a deterministic funnel that explains where users are in the product journey.

6. Deterministic User Funnel

Notebook cells: 41-46

Output to point at

Cell 42: funnel distribution table. Cell 43: Fig 3 funnel chart. Cell 45: Fig 11 transition matrix. Cell 46: Fig 12 time-to-stage.

What to say

The funnel assigns every user to exactly one stage based on observable rules: New, Exploring, Builder, AI User, Power User, Upgraded, or At Risk. This is not a model; it is deterministic logic that an engineer could implement.

What can be inferred

A large share of users are New, Exploring, or At Risk. The transition matrix shows how users move over 30 days, and the time-to-stage chart shows that important milestones often happen very early.

Transition line

With the model and funnel in place, I use them to produce product insights and recommended actions.

7. Behavioral Insight 1: AI Usage

Notebook cells: 47-50

Output to point at

Cell 49: mean and median comparison table. Cell 50: Fig 4 AI usage vs upgrade outcome.

What to say

The first insight is that AI power users are much more likely to upgrade. Upgraders average far more AI generations and more credit-pressure events than non-upgraders.

What can be inferred

This suggests that AI usage creates perceived value, and credit limits create a natural upgrade moment. A good action is a contextual upgrade nudge inside the AI workflow when usage or token pressure is high.

Transition line

The next insight looks at an even stronger production-intent signal: deployment.

8. Behavioral Insight 2: Deployment Intent

Notebook cells: 51-52

Output to point at

Cell 52: upgrade rate by deployment and AI usage flags.

What to say

Deployment is a strong signal because it means the user is moving from exploration into production or sharing. The notebook shows deployers upgrade at a much higher rate than non-deployers.

What can be inferred

The inference is that deployment should trigger a different kind of message than generic AI usage. The pitch should focus on reliability, production capacity, and collaboration.

Transition line

The final behavioral insight looks at how quickly users reach value after signup.

9. Behavioral Insight 3: Early Activation

Notebook cells: 53-54

Output to point at

Cell 54: upgrade rate by time-to-first-AI window, plus AI vs non-AI upgrade rate.

What to say

This section compares users based on when they first use AI. The bigger point is that users who reach meaningful product value are more likely to convert than users who never use AI.

What can be inferred

The product implication is that onboarding should push users toward a first AI generation quickly, ideally in the first session or first few days.

Transition line

I finish the demo by showing supporting visuals and the submission checklist.

10. Supporting Visuals and Wrap-Up

Notebook cells: 55-63

Output to point at

Cell 56: Fig 5 daily trend. Cell 58: Fig 6 retention. Cell 60: Fig 7 segmentation.

Cell 62: Fig 8 distribution. Cell 63: checklist.

What to say

These final figures support the overall story: product activity changes over time, retention drops by cohort, some personas convert better than others, and upgraders tend to be more active.

What can be inferred

The final checklist ties the work back to the challenge requirements: leakage-safe prediction, interpretable model, actionable tiers, deterministic funnel, transition logic, and behavioral insights.

Transition line

My closing line: the project turns raw event telemetry into a practical growth system: score users, explain the score, place them in a funnel stage, and recommend the next action.

Quick Reference: Figures and Cells

Use this when you need to jump during recording.

Fig 1	Cell 35	Model performance: ROC and precision-recall curves.
Fig 2	Cell 36	Feature importance: top XGBoost features.
Fig 9a/b	Cell 38	SHAP beeswarm and global SHAP bar chart.
Fig 10	Cell 40	Actionable upgrade risk tiers.
Fig 3	Cell 43	Funnel distribution.
Fig 11	Cell 45	Thirty-day stage transition matrix.
Fig 12	Cell 46	Median time-to-stage progression.
Fig 4	Cell 50	AI usage and upgrade outcome comparison.
Fig 5	Cell 56	Daily event-volume trend.
Fig 6	Cell 58	Retention cohort heatmap.
Fig 7	Cell 60	Persona segmentation.
Fig 8	Cell 62	Events-per-user distribution.